

# Key Component List

The parts whose numbers feed the software, the parameters and the simulator, with where each number came from

|          |                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------|
| Document | BOATY-KCL-001                                                                                        |
| Issue    | Issue D (baselined, sdr-baseline-1)                                                                  |
| Date     | 30 September 2026                                                                                    |
| Status   | Baselined at the System Design Review (tag sdr-baseline-1); changes only by change request           |
| Basis    | ADD Issue H, ICD Issue H, FMEA Issue G, SSS (HUL B, PRP C, PWR D, HLM F, MCP F, MCN F, REC B, SIM G) |
| Content  | 17 key components, 11 findings, 19 datasheet entries                                                 |

# Document control

| Issue | Date              | Change                                                                                                                                                                                                                                                                                                                                         | By                     |
|-------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| A     | 28 September 2026 | First issue.                                                                                                                                                                                                                                                                                                                                   | Claude (drafted)       |
| B     | 29 September 2026 | CR-04 accepted: SpeedyBee F405 WING APP is the baseline. Section 10 now records where each change went.                                                                                                                                                                                                                                        | Claude, owner decision |
| C     | 30 September 2026 | CR-08: the conditional antenna pole kit is outside the cap, in the field-kit budget.                                                                                                                                                                                                                                                           | Claude, owner decision |
| D     | 30 September 2026 | SDR RID-09: ArduPilot sources re-archived from the Rover-4.7.1 release commit db79216, the one the simulator is built from (was a11f7351). Differences: ALLOW_ARM_NO_COMPASS in both board files; a typo in the unused motorboat.parm. The build checks the commit matches the software. Status and basis from the baseline register (RID-08). | Claude                 |

| Role                | Name                      | Signature / date |
|---------------------|---------------------------|------------------|
| Author              | Claude, for Project Boaty |                  |
| Reviewer / approver | Project owner             |                  |

Review guidance: check the proposed parts against what you can actually buy, and the cost estimates against real prices. Every value is tagged with its source; orange-tagged values are guesses until measured.

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# 1. Purpose and how to read this list

The simulator and the software need exact numbers: which UART the mission computer is on, which ADC pin senses the motor rail, how fast an ESC stops when its signal disappears, how much thrust a pod gives. Until now the specifications said 'F405 class' and 'M10 GNSS'. This list turns those into specific parts and records the numbers that matter from each.

A **key component** is one whose characteristics feed the software, the ArduPilot parameters, the simulator's physics or an interface definition. Foam, glands and fasteners are not key components and stay in the ADD's BOM.

## 1.1 Where the numbers come from

Each value carries a source tag. The helm's numbers come mostly from ArduPilot's own board definition: the firmware we flash is compiled from those files, so they're more authoritative than a vendor PDF.

| Tag  | Meaning                                                                                                                                                                                                                                                 | Trust   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AP   | <b>ArduPilot source.</b> Read from ArduPilot's board definition, driver or parameter source at the commit below, and archived in docs/kcl/sources/. The firmware we flash is built from these files, so for the helm this beats the vendor's datasheet. | High    |
| FW   | <b>Firmware source.</b> Read from the ESC firmware (AM32) source at the commit below. Excerpt archived.                                                                                                                                                 | High    |
| WEB  | <b>Vendor spec (web).</b> Vendor or retailer specification seen through web search. Not yet checked against an archived PDF.                                                                                                                            | Medium  |
| KN   | <b>Known datasheet value.</b> A well-known datasheet figure quoted from memory. Must be checked against the archived PDF before it is relied on.                                                                                                        | Medium  |
| EST  | <b>Engineering estimate.</b> Calculated or assumed. Replaced by a measurement at the stage shown.                                                                                                                                                       | Low     |
| TEST | <b>Measure.</b> No trustworthy datasheet exists (salvaged, generic or water-loaded parts). The bench test is the datasheet.                                                                                                                             | Pending |

Vendor sites (and ardupilot.org) were not reachable from the environment this list was written in; GitHub was. So ArduPilot and AM32 sources are archived in docs/kcl/sources/, and vendor datasheets are listed in section 9 with a script to fetch them.

## 2. Findings

**CR-04 accepted (Issue B).** The baseline flight controller (Matek F405-TE) needed 9 V, and our 3S Li-ion pack sags below that under load near empty. The boat could have lost its autopilot during the very return-home that the low-battery failsafe starts. The SpeedyBee F405 WING APP runs from 7 V, keeps microSD logging, costs about £17 less, and is now the baseline (ADD DD-19). Section 5 has the numbers.

| ID    | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Type     | Goes to                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------|
| KF-01 | <b>The baseline flight controller browns out near empty.</b> The Matek F405-TE needs at least 9 V in. Our 3S Li-ion pack is at 9.6 V (resting) when the critical battery failsafe fires, and it sags under load: with acceptance-limit cells the pack resistance is about 0.22 $\Omega$ . So the F405-TE would reset during the return-home it is flying, above about 1.4 A of motor current (section 5). PWR-D12 already requires supplies to hold at 9.0 V, which the F405-TE cannot meet. The SpeedyBee F405 WING APP runs from 7 V, has microSD logging, 12 DShot outputs and an ArduPilot board definition, and costs about £17 less. | Decision | CR-04 accepted (Issue B): KC-01 is the SpeedyBee F405 WING APP (ADD DD-19) |
| KF-02 | <b>Limit motor power so sag can't brown out anything.</b> Even with the SpeedyBee, full thrust on a nearly flat, high-resistance pack could pull the rail below the ESCs' 7.2 V minimum. ArduPilot's BATT_WATT_MAX (Rover) reduces maximum throttle when power exceeds a limit. Set to 70 W, it keeps the rail above 7.5 V at the critical failsafe voltage while still leaving about 2 N of thrust, against 1.1 N needed for ENV-002.                                                                                                                                                                                                     | Change   | HLM parameter baseline (section 7)                                         |
| KF-03 | <b>The arming-key MOSFET must switch the positive side.</b> ICD IF-06 says 'gate pulled down → rail off', which reads like a low-side N-MOSFET, the kind of cheap trigger module sold for this. Switching the ESC negative while its signal ground is tied to the flight controller lets motor current find a way back through the signal wires, which can destroy the FC's output pins. So the key switch shall be a high-side P-MOSFET, with a gate RC to soft-start the ESC input capacitors (2 × 470 $\mu$ F).                                                                                                                         | Change   | ICD IF-06, SSS-PWR PWR-D07                                                 |
| KF-04 | <b>The flight controller's built-in wireless is a second, uncontrolled command path.</b> The SpeedyBee stack includes a Wi-Fi/Bluetooth board wired to UART6 and set up for MAVLink by default. That would let any phone near the boat talk to the autopilot, bypassing IF-01/IF-02 and the adult PIN. Don't fit the wireless board, and set SERIAL6_PROTOCOL = -1 as well.                                                                                                                                                                                                                                                                | Change   | HLM parameter baseline, ICD IF-02                                          |
| KF-05 | <b>Two microSD cards, not one.</b> The Pi Zero 2W boots from one card; the flight controller logs to another (DD-15). The ADD BOM counted one. +£4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Change   | ADD BOM                                                                    |
| KF-06 | <b>The boat link must be 2.4 GHz.</b> The Pi Zero 2W's radio is 2.4 GHz only. The bank access point must run on 2.4 GHz (channel 1, 6 or 11). MT7610U/MT7612U USB adapters have in-kernel (mt76) drivers with access-point mode on Raspberry Pi OS, so no out-of-tree driver is needed.                                                                                                                                                                                                                                                                                                                                                    | Change   | ICD IF-01                                                                  |
| KF-07 | <b>The simulator needs our own boat physics.</b> ArduPilot's built-in motorboat model gives 50 N at full throttle (hull drag equal to thrust at 10 m/s). Ours is roughly 4 N at 1.5 m/s. Simulated failsafe and weed tests would be meaningless at that scale. SITL has a JSON interface for an external physics model, so we write a small Python boat model parameterised from this list (section 8).                                                                                                                                                                                                                                    | Change   | Proposed DD-19; SSS-SIM, ICD IF-21                                         |
| KF-08 | <b>Turn off the ESC's own low-voltage cut-off.</b> AM32 has a configurable low-voltage cut-off that stops the motor after 10 s below its threshold. On a Li-ion pack it could strand the boat while ArduPilot is still trying to bring it home. Disable it in the ESC settings; battery protection belongs to the helm (FS-001) and the BMS.                                                                                                                                                                                                                                                                                               | Change   | SSS-PRP ESC configuration                                                  |
| KF-09 | <b>Free motor-RPM telemetry (opportunity).</b> AM32 supports bidirectional DShot, which reports each motor's RPM to ArduPilot. That gives B7 a direct 'motor dead / prop fouled' signal instead of inferring it from track divergence (FM-16, FM-27). It needs the motors on outputs 1 and 4, the only BIDIR-capable output on each of the first two timer groups. Not in the baseline; worth a bench trial.                                                                                                                                                                                                                               | Note     | Candidate for HLM Issue D                                                  |
| KF-10 | <b>Calibrate the battery monitor on the bench.</b> ArduPilot's board file and its README disagree on the voltage scale (11.05 against 11.5), which is a 4% difference and outside the $\pm 1\%$ of PWR-D13. The current scale (50 A/V) depends on the PDB fitted. Both get calibrated against a multimeter (IF-07 test).                                                                                                                                                                                                                                                                                                                   | Note     | IF-07 bench test                                                           |

| ID    | Finding                                                                                                                 | Type   | Goes to        |
|-------|-------------------------------------------------------------------------------------------------------------------------|--------|----------------|
| KF-11 | <b>The Pi 5 has no analogue audio output.</b> The speaker must be USB (or a USB sound card), not a 3.5 mm jack speaker. | Change | MCN purchasing |

### 3. The list

KC-08 and KC-13 are not used: the main switch and the moisture traces turned out not to feed any software value beyond what the specifications already say.

| ID    | Component                               | Qty | Proposed part                                                                                       | Status                    |
|-------|-----------------------------------------|-----|-----------------------------------------------------------------------------------------------------|---------------------------|
| KC-01 | Flight controller (helm) (HLM)          | 1   | SpeedyBee F405 WING APP (FC + PDB; wireless board not fitted)                                       | Baseline (CR-04 accepted) |
| KC-02 | GNSS receiver + compass (HLM)           | 1   | M10 GNSS with QMC5883L compass on one board (e.g. Matek/UMT M10Q-5883 or Foxeer M10Q-250-5883)      | Proposed                  |
| KC-03 | Electronic speed controller (×2) (PRP)  | 2   | AM32 single ESC, 20 A, 2-6S (e.g. OddityRC AM32 6S 20A Nano)                                        | Proposed                  |
| KC-04 | Thruster motor (×2) (PRP)               | 2   | 2205-2207 FPV outrunner, 1700-2300 KV, 5 mm shaft, run flooded                                      | Class only                |
| KC-05 | Battery cells (×3) (PWR)                | 3   | Salvaged 18650 Li-ion accepted to PWR-D20                                                           | Baseline (CR-03)          |
| KC-06 | Battery management system (PWR)         | 1   | 3S Li-ion BMS, 20-25 A, with balancing                                                              | Class only                |
| KC-07 | Arming-key switch (PWR)                 | 1   | Glass reed switch (normally open) driving a high-side P-MOSFET (AOD4185 class) on a small perfboard | Proposed                  |
| KC-09 | 5 V buck for the mission computer (PWR) | 1   | Synchronous buck module, 5.1 V, $\geq 3$ A continuous                                               | Class only                |
| KC-10 | Mission computer (MCP)                  | 1   | Raspberry Pi Zero 2 W                                                                               | Baseline                  |
| KC-11 | Camera (MCP)                            | 1   | OV5647 5 MP module with Pi Zero ribbon (Camera Module v1 compatible)                                | Baseline                  |
| KC-12 | Box temperature sensor (MCP)            | 1   | DS18B20 (TO-92)                                                                                     | Baseline (FMEA A-11)      |
| KC-14 | LED beacon (REC)                        | 1   | WS2812B addressable LEDs (4-8) on the FC LED pad                                                    | Baseline                  |
| KC-15 | Bank Wi-Fi adapter (MCN)                | 1   | USB adapter on MT7610U or MT7612U chipset with external antenna                                     | Proposed                  |
| KC-16 | Mission Control computer (MCN)          | 1   | Raspberry Pi 5 (owned)                                                                              | Baseline                  |
| KC-17 | Microphone and speaker (MCN)            | 1   | USB microphone (16 kHz mono capable) and USB speaker                                                | Class only                |
| KC-18 | Panel buttons (×4) (MCN)                | 4   | 30 mm arcade buttons with 5 V LED                                                                   | Class only                |
| KC-19 | microSD cards (×2) (MCP)                | 2   | 32 GB A1/U1 (one for the Pi Zero, one for the flight controller)                                    | Proposed (KF-05)          |

Motor rail is live only with the key in; the FC and mission computer stay powered with it out.

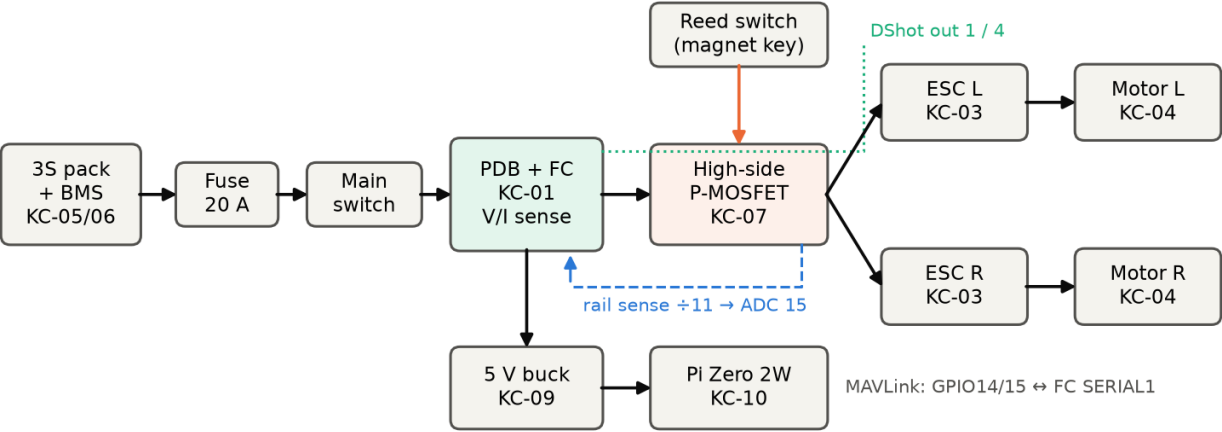


Figure 1. Boat power path with the proposed parts. The arming key switches the positive side of the motor rail (KF-03), and the helm senses that rail on its airspeed ADC pad.

## 4. Cost reconciliation

With the proposed parts the estimate is **£181**, against the £185 cap and £180 target (CON-001). Cheaper flight controller, realistic ESC price and a second microSD card roughly cancel out. The conditional antenna pole kit sits in the field-kit budget, outside the cap (CR-08).

| Group                                | Item                                                  | £ (est.)   | KC           | Change from ADD Issue E                   |
|--------------------------------------|-------------------------------------------------------|------------|--------------|-------------------------------------------|
| Boat                                 | Flight controller: SpeedyBee F405 WING APP (FC + PDB) | 45         | KC-01        | –£17 (was Matek F405-TE £62; CR-04)       |
| Boat                                 | M10 GNSS + QMC5883L compass                           | 14         | KC-02        | -                                         |
| Boat                                 | Pi Zero 2W                                            | 15         | KC-10        | -                                         |
| Boat                                 | OV5647 camera + Zero cable                            | 8          | KC-11        | -                                         |
| Boat                                 | microSD 32 GB × 2                                     | 8          | KC-19        | +£4 (KF-05)                               |
| Boat                                 | 2 × 2205-class motor                                  | 12         | KC-04        | -                                         |
| Boat                                 | 2 × AM32 20 A ESC                                     | 22         | KC-03        | +£12 (was £10; realistic price)           |
| Boat                                 | 3 × tested salvaged 18650 + 3S BMS                    | 7          | KC-05, KC-06 | -                                         |
| Boat                                 | Box, PG7 glands, fuse                                 | 6          | -            | -                                         |
| Boat                                 | Foam, hi-vis, WS2812B beacon                          | 5          | KC-14        | -                                         |
| Boat                                 | 5 V 3 A buck                                          | 3          | KC-09        | -                                         |
| Boat                                 | Reed switch + high-side P-MOSFET switch               | 4          | KC-07        | Same cost, different circuit (KF-03)      |
| Boat                                 | IP67 main power switch                                | 4          | -            | -                                         |
| Boat                                 | DS18B20 box temperature sensor                        | 2          | KC-12        | -                                         |
| Bank                                 | 4 arcade buttons (5 V LED)                            | 6          | KC-18        | -                                         |
| Bank                                 | USB mic + USB speaker                                 | 8          | KC-17        | Speaker must be USB (KF-11)               |
| Bank                                 | USB Wi-Fi adapter, MT7610U/MT7612U                    | 12         | KC-15        | -                                         |
| <b>Total (excluding conditional)</b> |                                                       | <b>181</b> |              | ADD Issue F baseline £181 (Issue E: £182) |
| Bank*                                | 2 m pole + 3 m USB extension (only if V-08 needs it)  | 8          | -            | Unchanged                                 |

Prices are estimates from search results and earlier work; UK retail sites couldn't be opened from here. Confirm at order time. If the ESCs come in dear, BLHeli\_S ESCs with Bluejay firmware save about £8.

## 5. Supply sag analysis

Pack resistance at the acceptance limits:  $3 \times 60 \text{ m}\Omega$  cells +  $15 \text{ m}\Omega$  BMS +  $25 \text{ m}\Omega$  fuse, switch and wiring = **0.22  $\Omega$** . The critical battery failsafe (BATT\_CRT\_VOLT) fires at 9.6 V resting, and the boat must then drive home on what's left.

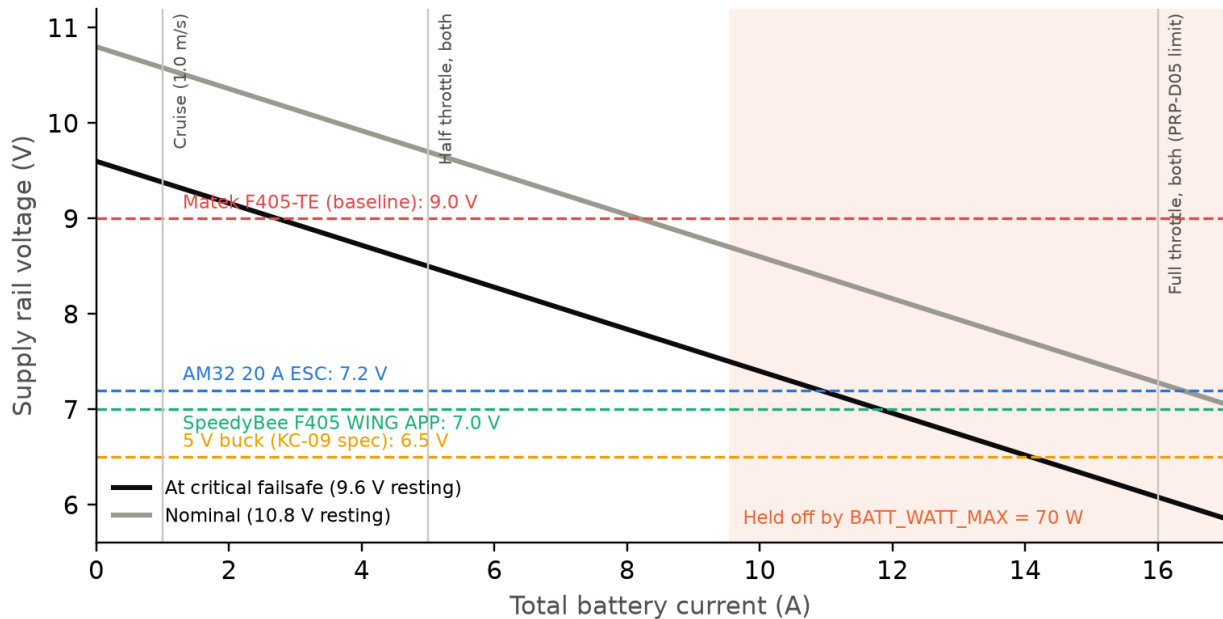


Figure 2. Rail voltage against battery current. The F405-TE (red) browns out above about 1.4 A near empty; cruise alone is about 1 A.

| Supply                   | Minimum input | Source | Max current at 9.6 V resting (+0.3 V margin) |
|--------------------------|---------------|--------|----------------------------------------------|
| Matek F405-TE (baseline) | 9.0 V         | WEB    | 1.4 A                                        |
| SpeedyBee F405 WING APP  | 7.0 V         | WEB    | 10.5 A                                       |
| AM32 20 A ESC            | 7.2 V         | WEB    | 9.5 A                                        |
| 5 V buck (KC-09 spec)    | 6.5 V         | EST    | 12.7 A                                       |

**Power limit.** BATT\_WATT\_MAX = 70 W keeps the rail above the highest of the remaining minimums (the ESC's 7.2 V) at the critical voltage. By momentum theory ( $T \propto P^{2/3}$ ) that still leaves about 2.1 N, against 1.1 N needed to make headway in the ENV-002 wind. On a full pack the limit rarely acts. Whether it is enough gets tested on L2 with a programmable load (PWR-D12).

## 6. Component data

### KC-01 Flight controller (helm)

|              |                                                                                           |  |  |
|--------------|-------------------------------------------------------------------------------------------|--|--|
| Part         | SpeedyBee F405 WING APP (FC + PDB; wireless board not fitted)                             |  |  |
| Alternatives | Baseline: Matek F405-TE (fails KF-01). Matek F405-WTE runs from 6.8 V but is end-of-life. |  |  |
| Why it's key | Every helm parameter, UART, ADC and output assignment comes from its board definition.    |  |  |
| Traces       | V-01, IF-04, IF-05, IF-07, IF-09, IF-22, DD-15, PWR-D12, PWR-D13                          |  |  |

| Parameter       | Value                                                                                                                                          | Src | Used by             |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------|
| MCU             | STM32F405, 168 MHz, 1 MB flash                                                                                                                 | AP  | V-01                |
| Input voltage   | 2-6S; 7-36 V                                                                                                                                   | WEB | PWR-D12, KF-01      |
| 5 V BEC         | 5.2 V, 2.4 A continuous / 3 A peak                                                                                                             | WEB | IF-07, IF-22, IF-09 |
| IMU / baro      | ICM42688-P (SPI) / SPL06 (I2C 0x76)                                                                                                            | AP  | HLM                 |
| Compass         | None on board; probes external I2C                                                                                                             | AP  | IF-22               |
| Logging         | microSD on SPI3 (FATFS); LOG_BACKEND_TYPE = 1                                                                                                  | AP  | LOG-001, DD-15      |
| UART map        | SERIAL1 USART1 (DMA) · SERIAL2 USART2 (SBUS inverter) · SERIAL3 USART3 (GPS) · SERIAL4 UART4 · SERIAL5 UART5 · SERIAL6 USART6 (wireless board) | AP  | IF-04, IF-22        |
| Companion UART  | SERIAL1 (USART1, DMA both ways): Pi Zero MAVLink                                                                                               | AP  | IF-04               |
| PWM outputs     | 12, all DShot. Groups 1-2 (TIM4), 3-4 (TIM3), 5-7 (TIM8), 8-10 (TIM2), 11-12 (TIM1). BIDIR on outputs 1 and 4. Output 12 = LED pad             | AP  | IF-05, IF-09        |
| Battery monitor | BATT_VOLT_PIN 10, BATT_CURR_PIN 11; scale 11.05 (hwdef) or 11.5 (README), 50 A/V: calibrate                                                    | AP  | IF-07, KF-10        |
| Spare ADC       | RSSI pad = ADC pin 14, AIRSPD pad = pin 15; 3.3 V full scale, no on-board divider                                                              | AP  | IF-06               |
| Current sensor  | 90 A continuous, 215 A peak                                                                                                                    | AP  | IF-07               |
| Mass            | ≈ 25 g FC + PDB (12 g allocated)                                                                                                               | EST | MEC-002             |

### KC-02 GNSS receiver + compass

|              |                                                                                                |  |  |
|--------------|------------------------------------------------------------------------------------------------|--|--|
| Part         | M10 GNSS with QMC5883L compass on one board (e.g. Matek/UMT M10Q-5883 or Foxeer M10Q-250-5883) |  |  |
| Alternatives | Any u-blox M10 + ArduPilot-supported compass (QMC5883L/P, IST8310)                             |  |  |
| Why it's key | Sets position noise, update rate and the compass driver, address and orientation parameters.   |  |  |
| Traces       | IF-22, NAV-008, V-16                                                                           |  |  |

| Parameter           | Value                                                                      | Src | Used by       |
|---------------------|----------------------------------------------------------------------------|-----|---------------|
| Receiver            | u-blox M10 (SAM-M10Q), GPS + Galileo + BeiDou + GLONASS                    | WEB | NAV-008       |
| Update rate         | 5 Hz UBX with four constellations; ArduPilot configures it (GPS_TYPE auto) | WEB | HLM           |
| Default baud        | 9600; ArduPilot auto-bauds and reconfigures                                | WEB | IF-22         |
| Horizontal accuracy | ≈ 1.5 m CEP open sky                                                       | KN  | SIM model     |
| Supply              | 4-9 V; ≈ 13 mA (receiver)                                                  | WEB | IF-22         |
| Compass             | QMC5883L on I2C, address 0x0D; ArduPilot probes it as external             | AP  | IF-22, TBC-09 |
| Temperature range   | −20 to 80 °C                                                               | WEB | -             |

### KC-03 Electronic speed controller (×2)

|              |                                                                                                                                       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Part         | AM32 single ESC, 20 A, 2-6S (e.g. OddityRC AM32 6S 20A Nano)                                                                          |
| Alternatives | BLHeli_S ESC flashed with Bluejay ( $\approx$ £7 each; SERVO_DSHOT_ESC = 2). AM32 preferred: maintained, configurable, RPM telemetry. |
| Why it's key | Its protocol, reversing, signal-loss and low-voltage behaviour are failsafe behaviour. They are modelled in SITL and tested on L2.    |
| Traces       | IF-05, PRP-D05, PRP-D06, PRP-D07, PRP-D08, V-09                                                                                       |

| Parameter           | Value                                                            | Src | Used by               |
|---------------------|------------------------------------------------------------------|-----|-----------------------|
| Protocol            | DSHOT300/600 and PWM                                             | WEB | IF-05                 |
| Reversible          | 3D mode via SERVO_BLM_3DMASK (ArduPilot configures AM32 at boot) | AP  | IF-05, PRP-D01        |
| Voltage             | 7.2-25.2 V (2-6S)                                                | WEB | KF-02                 |
| Current             | 20 A continuous, 30 A burst (10 s)                               | WEB | PRP-D05               |
| Signal loss, armed  | All phases off and reset after 0.5 s                             | FW  | PRP-D07, FS-008, V-09 |
| Re-arm              | Needs > 1 s of zero command before it will drive again           | FW  | PRP-D08               |
| Low-voltage cut-off | Configurable; stops after 10 s below threshold → disable (KF-08) | FW  | FS-001                |
| Telemetry           | Bidirectional DSHOT RPM (optional, KF-09)                        | WEB | FM-16                 |
| BEC                 | 5 V 1 A: leave unconnected (don't parallel with the FC BEC)      | WEB | IF-07                 |
| Mass                | 2.6 g board; 36 × 24 mm                                          | WEB | MEC-002               |

### KC-04 Thruster motor (×2)

|              |                                                                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Part         | 2205-2207 FPV outrunner, 1700-2300 KV, 5 mm shaft, run flooded                                                                                      |
| Alternatives | Sealed ROV thruster motors (much dearer; future option)                                                                                             |
| Why it's key | Thrust and current against throttle are the most important simulator inputs, and no datasheet covers a small motor turning a printed prop in water. |
| Traces       | PRP-D02, PRP-D05, IF-06                                                                                                                             |

| Parameter                | Value                                                      | Src  | Used by       |
|--------------------------|------------------------------------------------------------|------|---------------|
| Stator / KV              | 2205-2207 / 1700-2300 KV (class)                           | EST  | PRP-D02       |
| Static thrust, each      | $\geq 2.0$ N forward, $\geq 0.75$ N astern (PRP-D02 split) | EST  | SIM model     |
| Thrust vs throttle       | Assume $T \propto \text{throttle}^2$ until measured        | EST  | SIM model     |
| Current at full throttle | $\leq 8$ A each (PRP-D05 limit)                            | TEST | IF-06, TBC-06 |
| Spin-up time constant    | $\approx 0.1$ s                                            | EST  | SIM model     |

### KC-05 Battery cells (×3)

|              |                                                                                         |
|--------------|-----------------------------------------------------------------------------------------|
| Part         | Salvaged 18650 Li-ion accepted to PWR-D20                                               |
| Alternatives | New cells (e.g. 3000 mAh, 15 A class) if too few pass                                   |
| Why it's key | Capacity and internal resistance set the failsafe thresholds, endurance and supply sag. |
| Traces       | PWR-D01, PWR-D20, PWR-D21                                                               |

| Parameter           | Value                                                                | Src  | Used by                |
|---------------------|----------------------------------------------------------------------|------|------------------------|
| Capacity            | $\geq 2500$ mAh each at 1 A (acceptance)                             | TEST | PWR-D20, BATT_CAPACITY |
| Internal resistance | $\leq 60$ m $\Omega$ each (acceptance); matched within 10 m $\Omega$ | TEST | KF-01, KF-02           |
| Voltage             | 4.2 V full, 3.6 V nominal, 2.5-3.0 V empty per cell                  | KN   | FS-001                 |

| Parameter | Value                             | Src | Used by |
|-----------|-----------------------------------|-----|---------|
| Pack      | 3S1P: 12.6 V full, 10.8 V nominal | EST | IF-06   |

## KC-06 Battery management system

|              |                                                                                                      |
|--------------|------------------------------------------------------------------------------------------------------|
| Part         | 3S Li-ion BMS, 20-25 A, with balancing                                                               |
| Alternatives | -                                                                                                    |
| Why it's key | Its over-discharge cut-off is the last line after the helm failsafes; it must never act before them. |
| Traces       | PWR-D02                                                                                              |

| Parameter         | Value                                       | Src  | Used by |
|-------------------|---------------------------------------------|------|---------|
| Over-discharge    | 2.5-2.8 V/cell (PWR-D02)                    | TEST | FS-001  |
| Over-current trip | $\leq 25$ A                                 | TEST | PWR-D02 |
| Series resistance | $\approx 15$ m $\Omega$ (in the sag budget) | EST  | KF-01   |

## KC-07 Arming-key switch

|              |                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------|
| Part         | Glass reed switch (normally open) driving a high-side P-MOSFET (AOD4185 class) on a small perfboard            |
| Alternatives | Low-side trigger modules are not acceptable (KF-03)                                                            |
| Why it's key | Its state is what the helm senses (BATT2) to allow arming. Its switching time and inrush are part of L2 tests. |
| Traces       | IF-06, PWR-D07, PWR-D08, PWR-D09, V-13                                                                         |

| Parameter  | Value                                                                                                                                                              | Src | Used by       |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------|
| MOSFET     | P-channel, $V_{DS} \geq 30$ V, $I_D \geq 30$ A, $R_{DS(on)} \leq 15$ m $\Omega$ at $V_{GS} = -10$ V                                                                | EST | IF-06         |
| Gate drive | Reed closes $\rightarrow$ gate pulled toward ground via divider, $V_{GS}$ clamped $-10$ V; open $\rightarrow 10$ k $\Omega$ pulls gate to source (off, fails safe) | EST | PWR-D07       |
| Soft start | Gate RC $\approx 10$ ms to limit inrush into $2 \times 470$ $\mu$ F                                                                                                | EST | KF-03         |
| Rail sense | Divider 10 k $\Omega$ / 1 k $\Omega$ from the switched rail to the AIRSPD pad (ADC pin 15)                                                                         | EST | PWR-D08, V-13 |

## KC-09 5 V buck for the mission computer

|              |                                                                                                              |
|--------------|--------------------------------------------------------------------------------------------------------------|
| Part         | Synchronous buck module, 5.1 V, $\geq 3$ A continuous                                                        |
| Alternatives | Pololu D36V28F5 (dearer, well documented)                                                                    |
| Why it's key | Its minimum input voltage decides whether the mission computer survives sag. It is part of the sag analysis. |
| Traces       | IF-08, PWR-D11, PWR-D12                                                                                      |

| Parameter     | Value                                   | Src | Used by |
|---------------|-----------------------------------------|-----|---------|
| Output        | 5.1 V $\pm 2\%$ , $\geq 3$ A continuous | EST | IF-08   |
| Minimum input | $\leq 6.5$ V at 1 A load                | EST | KF-02   |
| Ripple        | $\leq 50$ mV p-p                        | EST | IF-08   |

## KC-10 Mission computer

|              |                                                                         |
|--------------|-------------------------------------------------------------------------|
| Part         | Raspberry Pi Zero 2 W                                                   |
| Alternatives | -                                                                       |
| Why it's key | Radio band, GPIO, 1-Wire and UART assignments feed IF-01, IF-03, IF-04. |
| Traces       | MCP-D01, MCP-D03, MCP-D05, IF-01, IF-04                                 |

| Parameter      | Value                                              | Src | Used by      |
|----------------|----------------------------------------------------|-----|--------------|
| SoC            | RP3A0: 4 × Cortex-A53 at 1 GHz, 512 MB             | KN  | MCP-D01      |
| Wi-Fi          | 2.4 GHz 802.11 b/g/n only                          | KN  | IF-01, KF-06 |
| UART to helm   | GPIO14/15 (PL011 with Bluetooth disabled), 3.3 V   | KN  | IF-04        |
| 1-Wire         | GPIO4 (w1-gpio overlay) for KC-12                  | KN  | MCP-D04      |
| Moisture input | GPIO with pull-up; traces short to ground when wet | EST | MCP-D03      |
| Power          | ≈ 0.6 W idle, ≈ 3 W peak with camera               | KN  | MCP-D05      |

## KC-11 Camera

|              |                                                                                   |
|--------------|-----------------------------------------------------------------------------------|
| Part         | OV5647 5 MP module with Pi Zero ribbon (Camera Module v1 compatible)              |
| Alternatives | Camera Module 3 (dearer, autofocus)                                               |
| Why it's key | Field of view, resolution and capture latency set B2's photo geometry and timing. |
| Traces       | MCP-D02, MCP-D10, MCP-D11                                                         |

| Parameter             | Value                                        | Src  | Used by |
|-----------------------|----------------------------------------------|------|---------|
| Sensor                | OV5647, 2592 × 1944; libcamera driver ov5647 | KN   | MCP-D10 |
| Horizontal FOV        | ≈ 54° (standard lens)                        | KN   | MCP-D02 |
| Still capture latency | ≈ 0.3-1 s in software                        | TEST | MCP-D10 |

## KC-12 Box temperature sensor

|              |                                                           |
|--------------|-----------------------------------------------------------|
| Part         | DS18B20 (TO-92)                                           |
| Alternatives | -                                                         |
| Why it's key | B6 raises RTL at 60 °C box temperature from this reading. |
| Traces       | A-11, MCP-D15                                             |

| Parameter       | Value                             | Src | Used by |
|-----------------|-----------------------------------|-----|---------|
| Accuracy        | ± 0.5 °C from −10 to +85 °C       | KN  | A-11    |
| Conversion time | 750 ms at 12-bit                  | KN  | MCP-D15 |
| Interface       | 1-Wire, 3.0-5.5 V, 4.7 kΩ pull-up | KN  | MCP-D04 |

## KC-14 LED beacon

|              |                                                                                          |
|--------------|------------------------------------------------------------------------------------------|
| Part         | WS2812B addressable LEDs (4-8) on the FC LED pad                                         |
| Alternatives | -                                                                                        |
| Why it's key | The helm drives it, so the notify-LED parameters and the output assignment depend on it. |
| Traces       | IF-09, REC-004                                                                           |

| Parameter     | Value                                                            | Src | Used by |
|---------------|------------------------------------------------------------------|-----|---------|
| Drive         | Output 12 (LED pad), SERVO12_FUNCTION = 120 (NeoPixel1)          | AP  | IF-09   |
| Supply / data | 5 V; 3.3 V data works on most WS2812B, marginal on some (TBC-08) | KN  | IF-09   |

| Parameter | Value                         | Src | Used by |
|-----------|-------------------------------|-----|---------|
| Current   | ≈ 60 mA per LED at full white | KN  | IF-07   |

## KC-15 Bank Wi-Fi adapter

|              |                                                                     |
|--------------|---------------------------------------------------------------------|
| Part         | USB adapter on MT7610U or MT7612U chipset with external antenna     |
| Alternatives | Pi 5 on-board Wi-Fi (shorter range; no external antenna)            |
| Why it's key | It must run as an access point on 2.4 GHz with an in-kernel driver. |
| Traces       | IF-01, MCP-D16                                                      |

| Parameter | Value                                         | Src | Used by |
|-----------|-----------------------------------------------|-----|---------|
| Driver    | mt76x0u / mt76x2u, in-kernel since Linux 4.19 | WEB | IF-01   |
| AP mode   | Supported (hostapd)                           | WEB | IF-01   |
| Band used | 2.4 GHz, channel 1/6/11 (Pi Zero 2W limit)    | KN  | KF-06   |

## KC-16 Mission Control computer

|              |                                                            |
|--------------|------------------------------------------------------------|
| Part         | Raspberry Pi 5 (owned)                                     |
| Alternatives | -                                                          |
| Why it's key | Hosts C1-C10. GPIO for the panel, USB for audio and Wi-Fi. |
| Traces       | IF-12                                                      |

| Parameter | Value                                            | Src | Used by |
|-----------|--------------------------------------------------|-----|---------|
| Audio     | No analogue output: USB speaker required (KF-11) | KN  | IF-12   |
| Power     | ≈ 5 W average, ≈ 8 W peak (STT)                  | EST | MC-006  |

## KC-17 Microphone and speaker

|              |                                                      |
|--------------|------------------------------------------------------|
| Part         | USB microphone (16 kHz mono capable) and USB speaker |
| Alternatives | USB sound card + small powered speaker               |
| Why it's key | On-device STT/TTS sample rate and device names.      |
| Traces       | NLI-002                                              |

| Parameter  | Value                                     | Src | Used by |
|------------|-------------------------------------------|-----|---------|
| Mic format | 16 kHz, 16-bit mono (resampled if needed) | EST | NLI-002 |
| Speaker    | USB audio class, ≥ 85 dB at 1 m           | EST | NLI-003 |

## KC-18 Panel buttons (×4)

|              |                                                  |
|--------------|--------------------------------------------------|
| Part         | 30 mm arcade buttons with 5 V LED                |
| Alternatives | 12 V-LED versions need a different driver: avoid |
| Why it's key | Debounce timing and LED drive feed IF-12.        |
| Traces       | IF-12, MC-002                                    |

| Parameter | Value                                     | Src | Used by |
|-----------|-------------------------------------------|-----|---------|
| Switch    | Normally-open microswitch; bounce ≤ 10 ms | EST | IF-12   |
| LED       | 5 V, ≈ 20 mA, driven through N-MOSFET     | EST | IF-12   |

## KC-19 microSD cards (×2)

|              |                                                                             |
|--------------|-----------------------------------------------------------------------------|
| Part         | 32 GB A1/U1 (one for the Pi Zero, one for the flight controller)            |
| Alternatives | -                                                                           |
| Why it's key | The FC needs FAT32 ( $\leq$ 32 GB). The Pi reserves $\geq$ 5 GB for photos. |
| Traces       | DD-15, MCP-D13                                                              |

| Parameter | Value                               | Src | Used by |
|-----------|-------------------------------------|-----|---------|
| FC card   | $\leq$ 32 GB, FAT32                 | AP  | DD-15   |
| Pi card   | 32 GB, A1 or better; read-only root | EST | MCP-D13 |

## 7. Helm allocation and parameter changes

From the SpeedyBeeF405WING board definition (DS-01). These close TBC-02 and most of TBC-05 once CR-04 is accepted.

| Function                 | FC resource                     | Notes                           |
|--------------------------|---------------------------------|---------------------------------|
| Mission computer (IF-04) | SERIAL1 = USART1 (DMA)          | Pads R1/T1; 115200 8N1          |
| GNSS (IF-22)             | SERIAL3 = USART3                | Board default GPS port          |
| Compass (IF-22)          | I2C1 (PB8/PB9)                  | QMC5883L at 0x0D                |
| Left motor (IF-05)       | Output 1 (TIM4, BIDIR)          | DShot300, 3D                    |
| Right motor (IF-05)      | Output 4 (TIM3, BIDIR)          | DShot300, 3D. Output 2/3 unused |
| Beacon (IF-09)           | Output 12 (LED pad, TIM1)       | WS2812B                         |
| Battery V/I (IF-07)      | ADC 10 / 11 via PDB             | Calibrate                       |
| Motor-rail sense (IF-06) | ADC 15 (AIRSPD pad)             | 10 kΩ / 1 kΩ divider            |
| Unused                   | SERIAL2, 4, 5; SERIAL6 disabled | Wireless board not fitted       |

### 7.1 Parameter baseline changes for SSS-HLM Issue D

| Parameter                                        | Proposed value                                | Was               | Src  | Refs         |
|--------------------------------------------------|-----------------------------------------------|-------------------|------|--------------|
| SERIAL1_PROTOCOL / SERIAL1_BAUD                  | 2 (MAVLink 2) / 115                           | SERIAL2 (TBC-02)  | AP   | KC-01, IF-04 |
| SERIAL2_PROTOCOL                                 | -1 (unused; SBUS inverter on RX)              | Companion         | AP   | KC-01        |
| SERIAL3_PROTOCOL                                 | 5 (GPS)                                       | -                 | AP   | KC-02, IF-22 |
| SERIAL6_PROTOCOL                                 | -1 (wireless board not fitted)                | Board default 2   | AP   | KF-04        |
| SERVO1_FUNCTION / SERVO4_FUNCTION                | 73 ThrottleLeft / 74 ThrottleRight            | SERVO1 / SERVO2   | AP   | KC-01, KF-09 |
| MOT_PWM_TYPE                                     | 6 (DShot300)                                  | DShot300 (TBC-05) | AP   | KC-03, IF-05 |
| SERVO_DSHOT_ESC                                  | 1 (AM32)                                      | -                 | AP   | KC-03        |
| SERVO_BLH_3DMASK                                 | 9 (outputs 1 and 4 reversible)                | -                 | AP   | KC-03, IF-05 |
| SERVO_BLH_RVMASK                                 | Set on the bench so forward is forward        | -                 | TEST | KC-03        |
| MOT_REV_DELAY                                    | 0.2 s (tune on L2)                            | -                 | EST  | KC-03, KC-04 |
| BATT_VOLT_MULT / BATT_AMP_PERVLT                 | Calibrated ( $\approx 11.05$ - $11.5$ / 50)   | Measured          | AP   | KF-10, IF-07 |
| BATT_FS_VOLTSRC                                  | 1 (sag-compensated)                           | -                 | AP   | KF-01        |
| BATT_WATT_MAX / MOT_BAT_WATT_TC                  | 70 W / 2 s                                    | -                 | EST  | KF-02        |
| BATT2_MONITOR / BATT2_VOLT_PIN / BATT2_VOLT_MULT | 3 (analogue voltage) / 15 (AIRSPD pad) / 11.0 | Analogue V        | AP   | KC-07, IF-06 |
| LOG_BACKEND_TYPE                                 | 1 (file on microSD)                           | File              | AP   | KC-01        |
| NTF_LED_TYPES / SERVO12_FUNCTION                 | NeoPixel / 120 (LED pad)                      | SERVO3 NeoPixel1  | AP   | KC-14, IF-09 |
| GPS_TYPE                                         | 1 (auto; u-blox configured by ArduPilot)      | Auto              | AP   | KC-02        |
| COMPASS_ORIENT / COMPASS_EXTERNAL                | Per mounting / 1                              | Per module        | AP   | KC-02, IF-22 |

## 8. Simulator model parameters

ArduPilot's built-in simulated motorboat gives 50 N at full throttle, for a much bigger boat (KF-07). The simulator will run the real ArduPilot firmware against a small Python boat model through SITL's JSON interface (DS-18). These are its starting parameters. Most are estimates, and the last column says which measurement replaces each one; the model is only as good as those measurements.

| Symbol                                     | Meaning                                                | Value                                               | Units                            | Source                 | Replaced by            |
|--------------------------------------------|--------------------------------------------------------|-----------------------------------------------------|----------------------------------|------------------------|------------------------|
| m                                          | Mass (all-up)                                          | 1.48                                                | kg                               | EST (ADD mass budget)  | Weigh the build        |
| m_a                                        | Surge added mass                                       | 10% of m                                            | kg                               | EST                    | Pool coast-down        |
| I_zz                                       | Yaw inertia, uniform box                               | 0.063                                               | kg·m <sup>2</sup>                | EST                    | Pool turn tests        |
| y_pod                                      | Pod lateral offset (HUL-D04 / 2)                       | ±0.135                                              | m                                | EST                    | CAD                    |
| T_fwd                                      | Max forward thrust, each pod                           | 2.0                                                 | N                                | EST (PRP-D02)          | L3 bollard pull        |
| T_ast                                      | Max astern thrust, each pod                            | 0.75                                                | N                                | EST (PRP-D02)          | L3 bollard pull        |
| T(u)                                       | Thrust law                                             | $T_{\max} \cdot u \cdot  u $                        | -                                | EST                    | Thrust stand           |
| $\tau_m$                                   | Motor spin-up time constant                            | 0.10                                                | s                                | EST                    | Thrust stand           |
| k_u                                        | Surge drag coefficient ( $F = k_u \cdot v \cdot  v $ ) | 1.62                                                | N·s <sup>2</sup> /m <sup>2</sup> | EST (SSS-PRP §6)       | Pool coast-down        |
| k_r                                        | Yaw damping ( $M = k_r \cdot r \cdot  r $ )            | 0.05                                                | N·m·s <sup>2</sup>               | EST                    | Pool turn tests        |
| A_w, C_d                                   | Windage area and drag coefficient                      | 0.035, 1.1                                          | m <sup>2</sup> , -               | EST (SSS-PRP §6)       | -                      |
| P(T)                                       | Electrical power per pod                               | $k_p \cdot T^{1.5}$ , $k_p$ set so cruise draws 8 W | W                                | EST                    | Thrust stand current   |
| P_hotel                                    | Hotel load                                             | 3.1                                                 | W                                | EST (ADD power budget) | Bench                  |
| C, R_pack                                  | Pack capacity and resistance                           | 2.5, 0.22                                           | Ah, $\Omega$                     | TEST (PWR-D20 limits)  | Cell acceptance data   |
| OCV(SoC)                                   | Open-circuit voltage curve                             | Generic NMC 18650 table                             | V                                | KN                     | Discharge test         |
| $\sigma_{\text{gnss}}$ , $f_{\text{gnss}}$ | GNSS noise and rate                                    | 1.0, 5                                              | m, Hz                            | KN / WEB               | Static log on the bank |
| V_min                                      | Supply brown-out thresholds                            | FC 7.0, ESC 7.2, buck 6.5                           | V                                | WEB / EST              | L2 sag test            |

A 3-DOF model is enough: surge, sway and yaw, with thrust from two pods, quadratic drag, windage, and the battery feeding back into available thrust. Waves, heel and pitch are left out: the catamaran is stiff (SSS-HUL section 6), and none of the SC tests depend on them.

## 9. Datasheet register

5 archived, 7 ready to fetch, the rest to locate or to measure. python3 docs/kcl/src/fetch\_datasheets.py downloads the 'To fetch' entries into docs/kcl/datasheets/ and records their SHA-256, so a datasheet can't silently change under us. Run it anywhere with ordinary internet access.

| ID    | KC    | Document                                                                                                                                                                                                                                                                                                                                                                                 | Status    |
|-------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| DS-01 | KC-01 | ArduPilot board definition, SpeedyBeeF405WING (hwdef.dat, README, defaults, pinout, wiring)<br><a href="https://github.com/ArduPilot/ardupilot/tree/dbe792162d06cab66c3475fd5556bf7a120f119e/libraries/AP_HAL_ChibiOS/hwdef/SpeedyBeeF405WING">https://github.com/ArduPilot/ardupilot/tree/dbe792162d06cab66c3475fd5556bf7a120f119e/libraries/AP_HAL_ChibiOS/hwdef/SpeedyBeeF405WING</a> | Archived  |
| DS-02 | KC-01 | SpeedyBee F405 WING APP user manual                                                                                                                                                                                                                                                                                                                                                      | To locate |
| DS-03 | KC-01 | ArduPilot board definition, MatekF405-TE, and the F405-TE family wiki page (for the comparison)<br><a href="https://github.com/ArduPilot/ardupilot/tree/dbe792162d06cab66c3475fd5556bf7a120f119e/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-TE">https://github.com/ArduPilot/ardupilot/tree/dbe792162d06cab66c3475fd5556bf7a120f119e/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-TE</a>       | Archived  |
| DS-04 | KC-02 | Matek M10Q-5883 product page and manual<br><a href="https://www.mateksys.com/?portfolio=m10q-5883">https://www.mateksys.com/?portfolio=m10q-5883</a>                                                                                                                                                                                                                                     | To fetch  |
| DS-05 | KC-02 | u-blox SAM-M10Q data sheet                                                                                                                                                                                                                                                                                                                                                               | To locate |
| DS-06 | KC-02 | QST QMC5883L data sheet                                                                                                                                                                                                                                                                                                                                                                  | To locate |
| DS-07 | KC-03 | OddityRC AM32 6S 20A Nano ESC product page<br><a href="https://oddityrc.com/products/am32-6s-20a-nano-esc">https://oddityrc.com/products/am32-6s-20a-nano-esc</a>                                                                                                                                                                                                                        | To fetch  |
| DS-08 | KC-03 | AM32 firmware source: signal-loss and re-arm logic (excerpt)<br><a href="https://github.com/am32-firmware/AM32/blob/2738df3240baa5bd4295b460cf0c5cfe0bd49d97/Src/main.c">https://github.com/am32-firmware/AM32/blob/2738df3240baa5bd4295b460cf0c5cfe0bd49d97/Src/main.c</a>                                                                                                              | Archived  |
| DS-09 | KC-04 | Motor + printed prop: thrust, current and RPM against throttle, in water                                                                                                                                                                                                                                                                                                                 | Measure   |
| DS-10 | KC-05 | Cell acceptance records (PWR-D21)                                                                                                                                                                                                                                                                                                                                                        | Measure   |
| DS-11 | KC-06 | BMS: trip thresholds measured on the bench                                                                                                                                                                                                                                                                                                                                               | Measure   |
| DS-12 | KC-07 | AOD4185 P-MOSFET data sheet                                                                                                                                                                                                                                                                                                                                                              | To locate |
| DS-13 | KC-10 | Raspberry Pi Zero 2 W product brief<br><a href="https://datasheets.raspberrypi.com/rpizero2/raspberry-pi-zero-2-w-product-brief.pdf">https://datasheets.raspberrypi.com/rpizero2/raspberry-pi-zero-2-w-product-brief.pdf</a>                                                                                                                                                             | To fetch  |
| DS-14 | KC-11 | Raspberry Pi camera documentation (OV5647 / Camera Module v1)<br><a href="https://www.raspberrypi.com/documentation/accessories/camera.html">https://www.raspberrypi.com/documentation/accessories/camera.html</a>                                                                                                                                                                       | To fetch  |
| DS-15 | KC-12 | Analog Devices DS18B20 data sheet<br><a href="https://www.analog.com/media/en/technical-documentation/data-sheets/DS18B20.pdf">https://www.analog.com/media/en/technical-documentation/data-sheets/DS18B20.pdf</a>                                                                                                                                                                       | To fetch  |
| DS-16 | KC-15 | MT7612U Linux support notes (morrownr/7612u)<br><a href="https://github.com/morrownr/7612u">https://github.com/morrownr/7612u</a>                                                                                                                                                                                                                                                        | To fetch  |
| DS-17 | KC-16 | Raspberry Pi 5 product brief<br><a href="https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-product-brief.pdf">https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-product-brief.pdf</a>                                                                                                                                                                                          | To fetch  |
| DS-18 | -     | ArduPilot SITL JSON physics interface (readme)<br><a href="https://github.com/ArduPilot/ardupilot/tree/dbe792162d06cab66c3475fd5556bf7a120f119e/libraries/SITL/examples/JSON">https://github.com/ArduPilot/ardupilot/tree/dbe792162d06cab66c3475fd5556bf7a120f119e/libraries/SITL/examples/JSON</a>                                                                                      | Archived  |
| DS-19 | -     | ArduPilot SITL motorboat and skid-steer default parameters<br><a href="https://github.com/ArduPilot/ardupilot/tree/dbe792162d06cab66c3475fd5556bf7a120f119e/Tools/autotest/default_params">https://github.com/ArduPilot/ardupilot/tree/dbe792162d06cab66c3475fd5556bf7a120f119e/Tools/autotest/default_params</a>                                                                        | Archived  |

'Measure' entries are deliberate: for salvaged cells, a generic BMS and a drone motor in water, no datasheet would be trustworthy. The bench result becomes the datasheet.

## 10. Where the changes went

| Document        | Where it went (Issue B)                                                                                                                                                                                                                                                          | Findings                   |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| ADD Issue F     | Done: KC-01 in HLM, BOM (£181) and mass budget (DD-19); own boat physics via SITL JSON (DD-20); DD-15 unchanged (microSD kept).                                                                                                                                                  | KF-01, KF-05, KF-07        |
| ICD Issue E     | Done: IF-04 on SERIAL1 (TBC-02 closed); IF-05 outputs 1 and 4, AM32 3D DShot (TBC-05 closed, bench check with V-09); IF-06 high-side switch and AIRSPD-pad sense (TBC-07 closed); SERIAL6 off; IF-21 rewritten for the JSON model (TBC-03/04 closed). IF-01 was already 2.4 GHz. | KF-03, KF-04, KF-06, KF-07 |
| SSS-HLM Issue D | Done: parameter baseline now generated from software/params; BATT_WATT_MAX (HLM-D43) and one command path (HLM-D44).                                                                                                                                                             | KF-02, KF-04               |
| SSS-PWR Issue C | Done: PWR-D07 high-side; PWR-D22 supplies work down to 7.5 V under the power limit.                                                                                                                                                                                              | KF-01, KF-03               |
| SSS-PRP Issue B | Done: AM32 ESC and its configuration (3D, LVC off, BEC unused; PRP-D16).                                                                                                                                                                                                         | KF-08                      |
| SSS-SIM Issue D | Done: boat model from this list (SIM-D25); injections on the model side (SIM-D07).                                                                                                                                                                                               | KF-07                      |
| FMEA Issue D    | Done: FM-08 (supply brown-out) re-rated with its real cause and closed by A-21. Still open: RPM telemetry as a detection control for FM-16/FM-50 (dead motor vs weed).                                                                                                           | KF-01, KF-09               |

### 10.1 Measurements that matter most to the simulator

- **Thrust, current and RPM against throttle** for one pod in water (DS-09). A luggage scale and a bucket will do. This single curve sets the model's thrust, power and endurance.
- **Cell capacity and resistance** from acceptance testing (DS-10). These set R\_pack and the sag margin.
- **ESC signal-loss stop time** on the bench (V-09). The AM32 source says 0.5 s; confirm on the actual ESC.
- **Coast-down and turn tests** in the pool to replace the drag and inertia estimates.

## Appendix A. Archived sources

| Source                                       | Repository and commit                   |
|----------------------------------------------|-----------------------------------------|
| ArduPilot firmware (board definitions, SITL) | ArduPilot/ardupilot @ dbe792162d06      |
| ArduPilot wiki (F405-TE family page)         | ArduPilot/ardupilot_wiki @ 5365bb696d91 |
| AM32 ESC firmware (signal-loss excerpt)      | am32-firmware/AM32 @ 2738df3240ba       |

ArduPilot and AM32 are GPLv3; the archived files are unmodified copies or excerpts, kept for reference.